

2 Core concepts of CSS

In this chapter we will take a closer look at the core concepts of basic CSS design. These concepts are required to understand how CSS works, the intrinsic working of styling and what happens behind the scenes when applying these styles. We will be looking into concepts like selectors, cascade, grouping, inheritance and measurements

2.1 Selectors (ID and Class)

In the previous examples we have only seen a few base selectors in use for CSS. A base selector is basically an existing HTML tag and using CSS you can redefine some of its default properties to style the whole element. For example, we have already seen how CSS changes the text colour of a paragraph using the selector 'p' which is a basic HTML tag.

In addition to these selectors, CSS allows you define your own custom selectors, namely the ID and class selectors. IDs and classes are applied to HTML elements as attributes providing more control over the presentation. It is common among novice developers to confuse ID with class. But understanding the differences will make your work a lot easier.

2.1.1 IDs

An ID is a unique identifier to an element and can only be used once per page. Typically, an ID is used for any unique page element such as the header, main navigation, footer, or other key parts in the page design.

The ID is applied to an HTML element by referencing it in the HTML using the `id="name"` attribute immediately after the opening tag within an element. To illustrate this let us work with two IDs named `highlight` and `default`, respectively, and by applying them to two paragraphs:

```
<p id="highlight">This paragraph has red text.</p>
<p id="default">This paragraph has dark gray text.</p>
```

The corresponding CSS uses the hash (#) character to denote the rule using an ID as a selector.

```
/* Define highlighted text */
#highlight {
color:#red;
}
```